

Syllabus for

Academic year 2021/2022 >

ERE091 - Automatic control

Reglerteknik

Syllabus adopted 2021-02-26 by Head of Programme (or corresponding)

Owner: [TKTFY](#)**4,5 Credits****Grading:** TH - Pass with distinction (5), Pass with credit (4), Pass (3), Fail**Education cycle:** First-cycle**Main field of study:** Automation and Mechatronics Engineering, Engineering Physics**Department:** 32 - ELECTRICAL ENGINEERING**Teaching language:** Swedish**Application code:** 57126**Open for exchange students:** No

Maximum participants: 100

Only students with the course round in the programme plan

Credit distribution

Module	Sp1	Sp2	Sp3	Sp4	Summer course 	No Sp	Examination dates 
0194 Examination 4,5 c Grading: TH				4,5 c			Fri 03/06-2022 pm J, Fri 08/10-2021 pm J  Fri 26/08-2022 am J

In programs

[TKTFY ENGINEERING PHYSICS, Year 2 \(compulsory elective\)](#)

Examiner:

[Bengt Lennartson](#)[Go to course homepage](#)

Eligibility

General entry requirements for bachelor's level (first cycle)

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Specific entry requirements

The same as for the programme that owns the course.

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Course specific prerequisites

Mathematical analysis in one and several variables. Basic theory of matrices, in particular eigenvalues. Complex numbers. Linear ordinary differential equations and transforms. Basic knowledge in mechanics, electricity as well as electronics for applications. Basic course in programming as a background for use of computer aids.

Aim

The purpose of this course is to introduce the concept of a dynamic system, and to demonstrate its application to different areas of technology. Yet another key concept is feedback and in particular the assessment of the stability of such a system. The course will teach theory and techniques for the design of both PID-controllers and state feedback controllers. Feedforward control will be addressed as well.

Learning outcomes (after completion of the course the student should be able to)

- Understand and explain the purpose of linear control systems and basic control terminology.
- Describe and explain the most important properties of linear dynamical systems.
- Formulate models as dynamic systems, frequently encountered in a technical context. Formulate models as transfer/frequency functions, as state equations.
- Transform in both directions between linear state equations and transfer/frequency functions, especially for single-input single-output systems. Linearize nonlinear state equations.
- Analyse feedback systems, emphasizing stability assessment based on the Nyquist criterion. Formulating

solutions to state equations, using state transition matrices.

- Define and explain the principle of P-, I-, PI-, PD- and PID-controllers in a control loop, as well as being able to carry out design for such controllers, in particular by use of Bode plot techniques.
- Analyse feedback systems, using sensitivity functions, particularly to estimate how large modelling errors a control system can handle without risking instability.
- Describe and explain the feedforward, cascade control, and dead time compensation.
- Explain and apply the concept of controllability and observability for the design of state feedback controllers and observers using the pole placement method.
- Discretize analog systems, explain the function of a computer control system, and explain the sampling technique.

Content

- The course is a fundamental course in system dynamics and control (linear systems). State-space models for linear and nonlinear systems is introduced. Linearization of state equations and obtaining transfer functions.
- Analysis of linear dynamical systems. Analysis of feedback systems. The Nyquist stability criterion. Root locus.
- P-, I-, PI-, PD- and PID-controllers and their most important properties. Bode diagrams. Nichols charts. Non-minimum phase systems. Design of control systems, particularly using compensation in frequency domain. Sensitivity functions and robustness.

Feedforward control, cascade control and dead time compensation.

- Linear state space methodology. Stability. State transition matrices. Controllability and observability. State feedback controllers and observers. Output feedback control.
- Computer controlled systems by sampling, and time discretization. Sampled transfer functions. A short introduction to linear quadratic control and Kalman filtering.

Organisation

The course is divided into a series of lectures, problem solving, and a mandatory project including laboratory sessions.

Literature

Textbook and lecture notes (Suggestions for literature will be given at the first lecture).

Examination including compulsory elements

Written exam with TH grading; Project with laboratory sessions (pass/fail).

The course examiner may assess individual students in other ways than what is stated above if there are special reasons for doing so, for example if a student has a decision from Chalmers on educational support due to disability.